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ISGF India Smart Grid Forum

India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

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Session : Role of Small Modular Reactors (SMR) in the Net Zero Power Systems

Presented By

Ganapati Myneni, Co-Founder, BSCE Systems, Inc

Projects further along with siting, vendor selection (financing at various stages)

Emerging SMR projects… now total >$140 billion in value to 2040

								Estimated	
				Operating by	MWs	[Reactors]	Total	$Billions	$Millions
		Projects			Average	Units	MWs	Total Cost	Cost per MW
		24		2028-2040	180	208	17,813	**$147.7 B**	$8.29 M
Locale	**Expected SMR Project**	**SMR Site**	**Reactor**	**First Operations**	**Size (MWe)**	**Number (Reactors)**	**Total MWs**	**Project Cost ($Billion)**	**Cost per MW ($Mil)**
USA, Utah	UAMPS	INL, ID	NuScale LWR		77	6	462	WITHDRAWN	
USA, several	NUCOR EAFs	Several: KY, WV	NuScale LWR		77	12	924	UNCERTAIN	
CAN, Ontario	OPG	Darlington, ON	GE LWR	2028	300	4	1200	$7.20 B	$6.0 M
UK Site-WYLFA	RR SMR	Wales, UK TBD	Rolls Royce	2032	470	2	940	$12.00 B	$12.8 M
UK Site	Hartlepool UK	Teesside, UK	X-energy	2035	80	12	960	$15.00 B	$15.6 M
CAN, NB	NB SMR	Pt. LePreau, NB	ARC-100 AMR		100	1	100	UNCERTAIN	
CAN, Ontario	CNL	Chalk River, ON	USNC Micro		5	1	5	WITHDRAWN	
USA, various	SWITCH	Data Centers	OKLO	2040	30	120	3600	UNCERTAIN	
USA DOE ARDP	TerraPower	Kemmerer, WY	Natrium AMR	2032	345	1	345	$5.00 B	$14.5 M
USA DOE ARDP	DOW Chem	Seadrift, TX	X-e 100	2032	80	4	320	$6.00 B	$18.8 M
USA DOE ARDP	X-energy	Hanford, WA	X-e 100	2032	80	12	960	$12.00 B	$12.5 M
Poland	PKN Orlen	Multiple sites	GE-BWRX	2030	300	6	1800	$15.00 B	$8.3 M
Poland	SYNTHOS	Oświęcim, PL	LWR SMR	2032	300	2	600	$4.00 B	$6.7 M
Estonia	Fermi Energia	Viru-Nigula, ES	GE-BWRX	2035	300	2	600	$8.00 B	$13.3 M
Czechia	CEZ	Temelin, CZ	Rolls Royce	2035	470	6	2820	$30.00 B	$10.6 M
Bulgaria	Industrial AMR	TBD	AMR	2035	80	2	160	$4.00 B	$25.0 M
Slovenia	Krško-2	Krško site, SLO	LWR SMR	2032	300	2	600	$8.00 B	$13.3 M
Romania	Cernavoda SMR	Doicești, ROM	NuScale LWR	2028	77	6	462	$8.00 B	$17.3 M
CHINA & RUSSIA									
CH, Shanghai	**China NNC AMR**	Shandong AMR	HTR-PM	**2024**	210	1	210	$3.00 B	$14.3 M
RU, Siberia	Seversk Chemical + Rosatom MOX	Seversk, RU	TVEL BREST Lead-cooled	2027	300	1	300	$4.00 B	$13.3 M
RU, Arctic Circle	**Arctic Port**	Pevek, RU	2 x KLT-40C	**2020**	35	2	70	$1.00 B	$14.3 M
CH, Hainan	**CNNC**	Changjiang	ACP100 PWR	2026	125	1	125	$1.50 B	$12.0 M
CH, Floating	CNNC	Floating SMR	ACP100 PWR	2028	125	1	125	$2.00 B	$16.0 M
CH, Jiangxi	CNNC	Ningdu	ACP100 PWR	2028	125	1	125	$2.00 B	$16.0 M

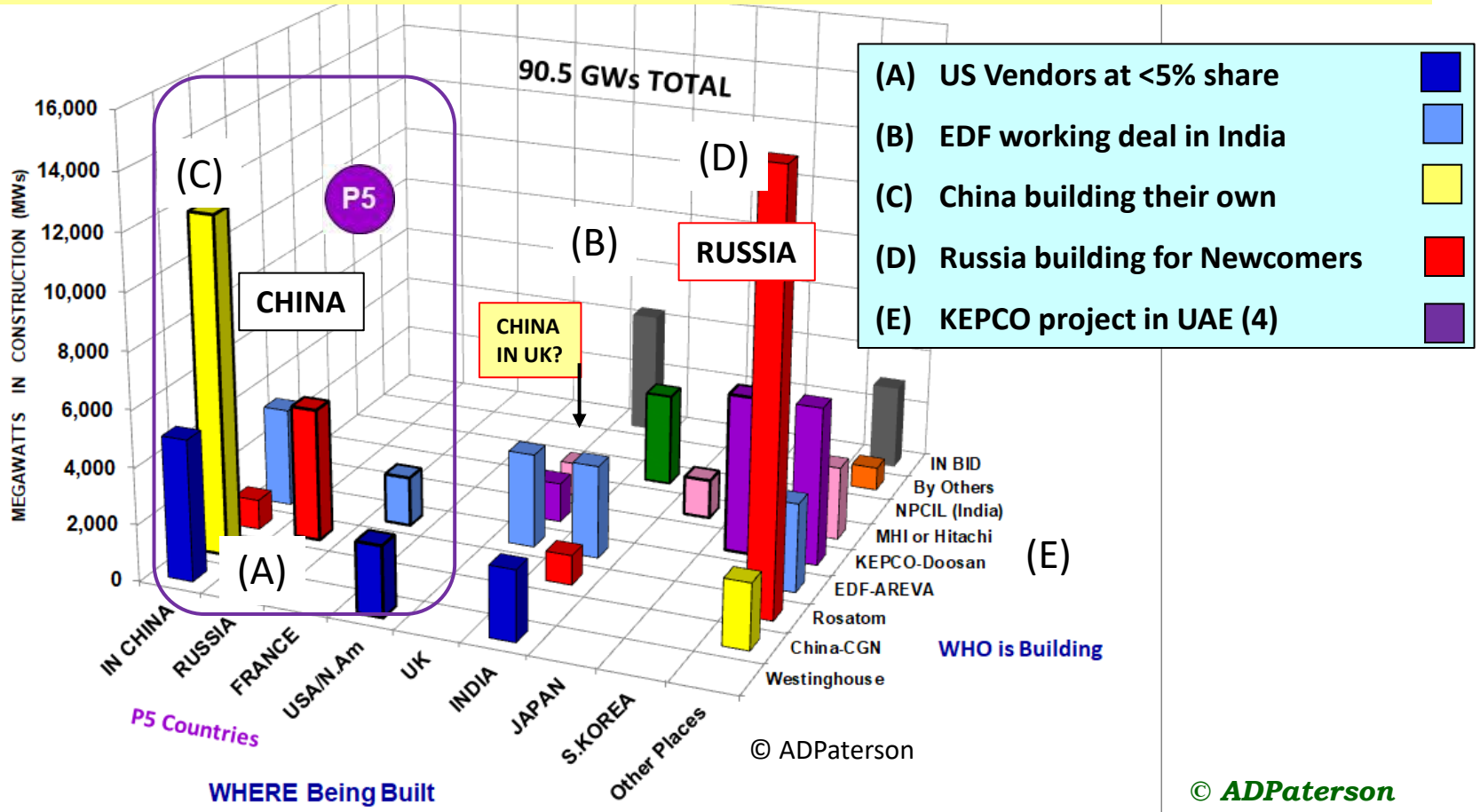
Standard LWR
Advanced unit

China or Russia	Stage of Development
Fuel load / Ops	5
Under Constr	4
Financing signed	3
Siting selected	2
JV Formed	1

Current Project Landscape: Who, Where, How much (GWs)

China & Russia most active on Nuclear Project Gameboard

From WNA listing of projects (2018-2025), augmented with trade press reports on projects very close to financing agreement. 30 GWs of 90 GWs being built captively.



WNA: www.world-nuclear.org/information-library/current-and-future-generation/plans-for-new-reactors-worldwide.aspx

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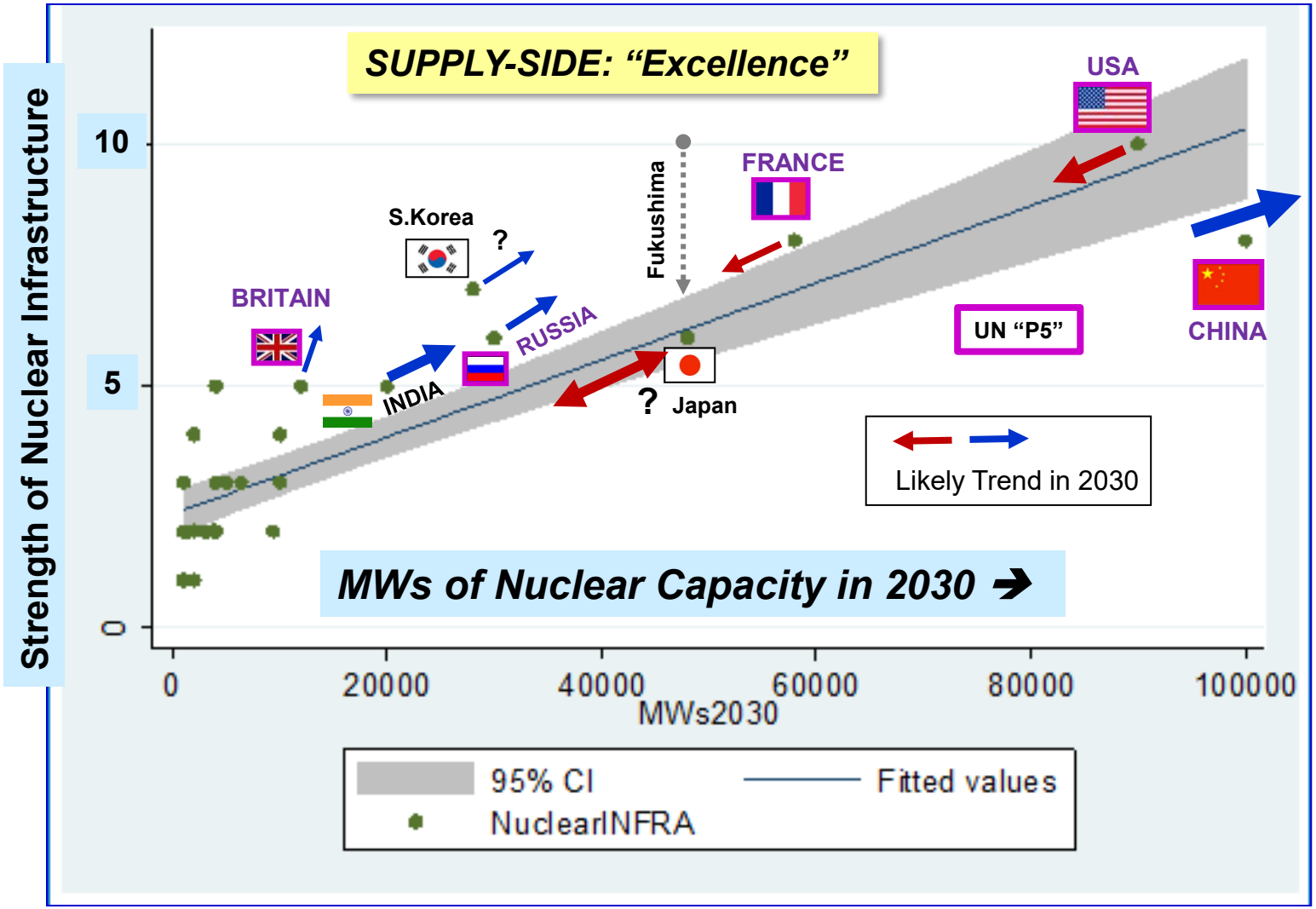
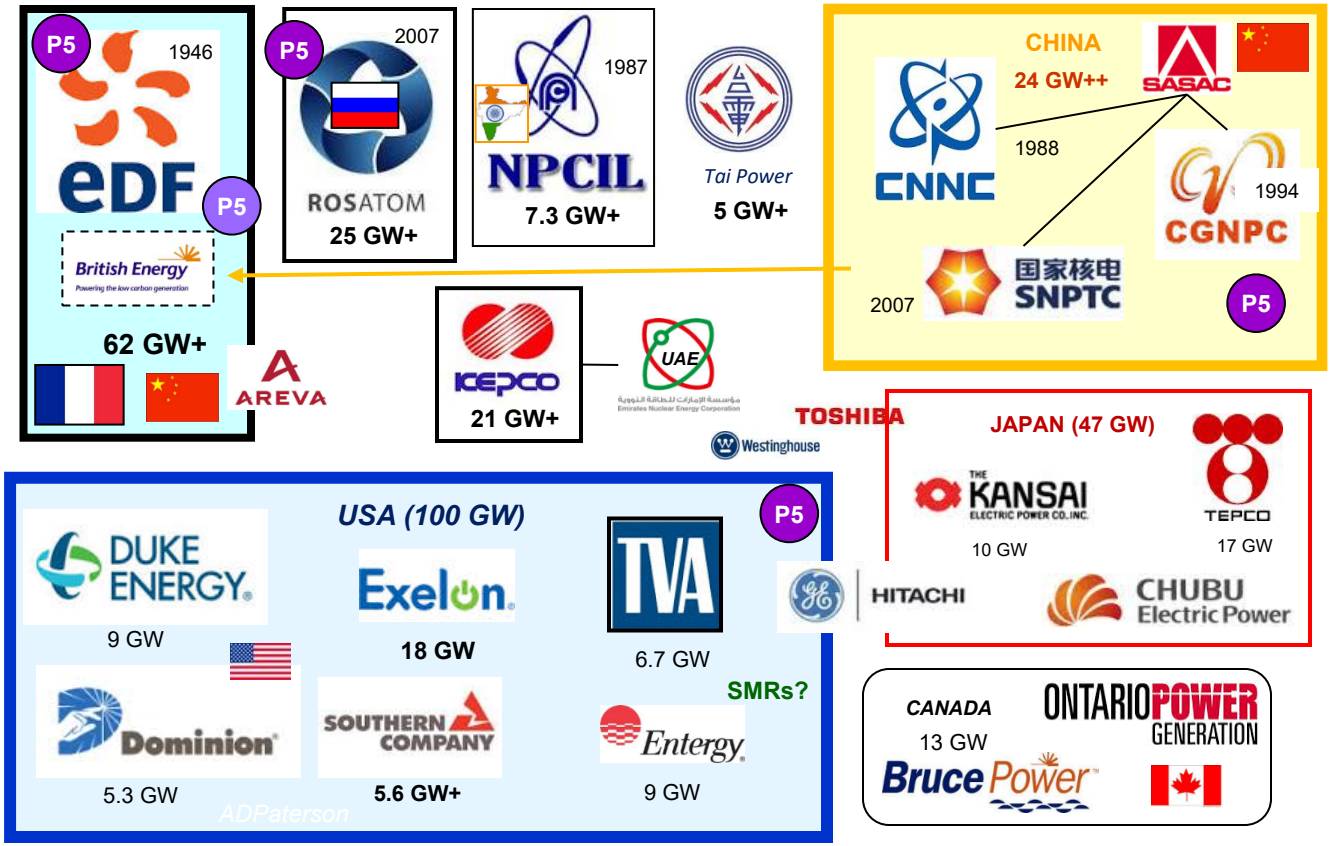
Leaders in Nuclear Infrastructure + MWs Planned for 2030



Ownership in US and Japan is more fragmented than in other countries

Global Nuclear Entities, P5 and "Next 5" in 2018



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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)